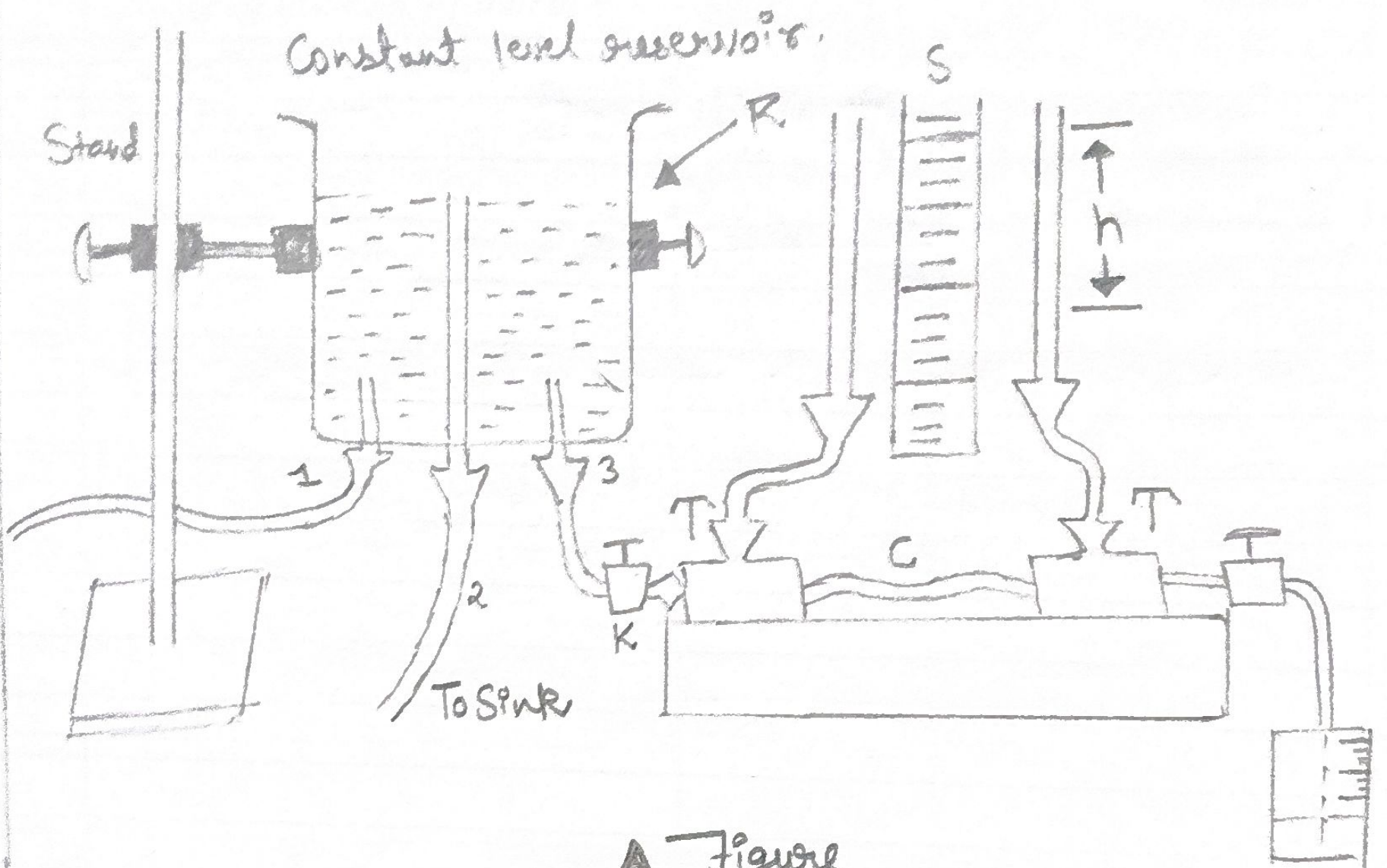
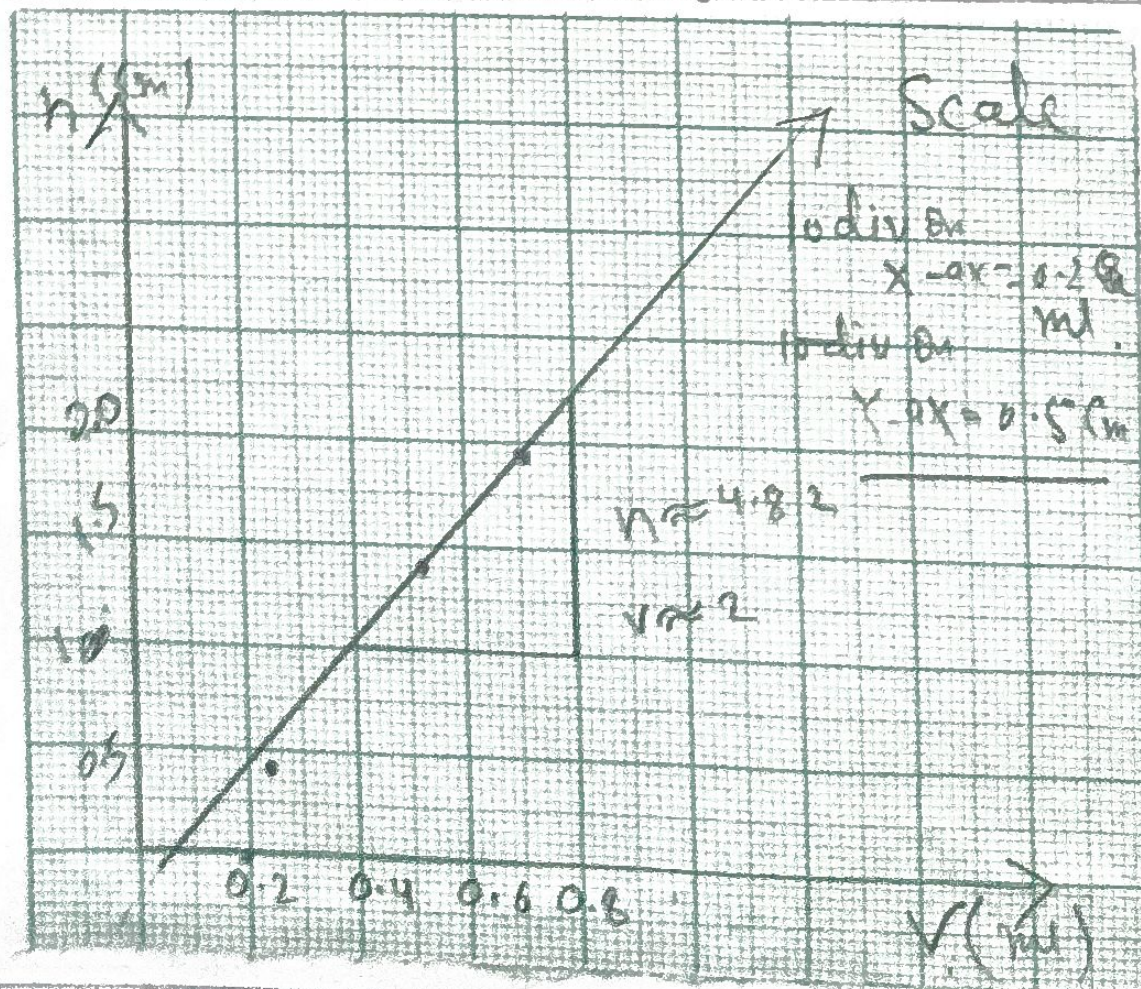




A Figure



Experiment No - 1

Object — To determine the Coefficient of viscosity of a liquid by Poiseuille's method

Apparatus Required —

Poiseuille's apparatus

A Capillary tube of uniform diameter

Graduated Cylinder

Cylinder

Stop watch

Manometer

Thermometer, stand

Formulae Used

When a liquid which wet, the surface is allowed to flow through a uniform capillary tube of length l & radius r , the volume V flowing out per second against a constant difference of pressure across its end, then coefficient of viscosity of a liquid is given by —

$$\eta = \frac{\pi P r^4}{8 V l}$$

$$(P = h \rho g)$$

Calculation — From Table —

$$\text{Average } [\eta/\nu] = \frac{1.7 + 2.47 + 3.01 + 2.47}{4} = 2.41$$

The Coefficient of viscosity of a liquid is given by —

$$\eta = \frac{\pi h \rho g r^4}{8 \nu l} \quad \text{g/c-s or poise.}$$

$$\eta = \frac{3.14 \times 1 \times 981 \times (0.123)^4 \times 2.41}{8 \times 25}$$

$$\eta = 0.00845 \quad \text{g/c-s or poise}$$

From Graph —

$$\eta = \frac{\pi \rho g r^4}{8 L} \times \text{Slope.}$$

$$\eta = \frac{3.14 \times 1 \times (0.123)^4}{8 \times 25} \times 2.41$$

$$\eta = 0.0085 \quad \text{g/c-s or poise}$$

$$\eta = \frac{\pi h \rho g r^4}{8 V l} \quad \text{g/c-s or poise}$$

This is known as Poiseuille's relation.

Where,

h = Difference of water levels in manometer limbs, ρ
 ρ = Density of water (1.0 gm/cm^3).

r = Radius of capillary tube ~~is~~ l = length of capillary tube.

V = Vol. of water collected per second, g = acceleration due to gravity (981 cm/s^2).

Observation Table.

S.No	Manometer		Difference [$h = h_1 - h_2$] (cm)	Measurement of Vol.		$V = Q/t$	h/V
	One end (h_1)	Other end (h_2)		Collected water (Q)	Time t (Sec)		
1	35.4	35	0.4	14	60	0.233	1.71
2	35.4	34	1.4	34	60	0.566	2.47
3	35.4	33.5	1.9	38	60	0.633	3.01
4	35.4	34	1.4	34	60	0.566	2.47

 $\frac{h}{V}$
2.47

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Result →

The coefficient of viscosity of water at temperature 30 degree centigrade is 0.0085 poise.

Standard result —Viscosity of water —

Temperature (deg Centigrade)	Viscosity (Poise).
10	0.01311
25	0.01000
30	0.00800
35	0.00600

Precautions — & Source of Errors.

1. The tube should be of uniform bore and should be placed horizontal.
2. The value of pressure difference should not be taken large otherwise the motion will not be streamline but turbulent.

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3. The radius should be measured very carefully and accurately to minimize the percentage error.

4. Value of water collected in graduated cylinder should be large.



Viva Voice Questions

Q₁ What is Viscosity?

A property of fluids due to which they, oppose relative motion b/w adjacent fluid layers.

Q₂ What is the direction of viscous force?

Tangential to the surface but opposite to the direction of relative motion.

Q₃ What is coefficient of viscosity?

It is a constant for fluid and is a measure of its viscosity property.

Q₄ State Stokes law.

For a small spherical body of radius r falling down a liquid of coefficient of viscosity η , a viscous force acts whose magnitude is $F = 6\pi\eta rV$.

Q5 What is terminal velocity?

It is maximum const. velocity with which a small spherical body may fall freely through a fluid.

Q6 On what factor terminal velocity depend?

Terminal velocity is directly proportional to the square of the radius of falling body and inversely proportional to the coefficient of viscosity of fluid.

Q7 Do air or gases also poses the property of viscosity. Yes, but coefficient of viscosity of gases is less than liquid.

Q8 Which is more viscous water or glycerin?

glycerin.

Q9 How does the coefficient of viscosity of a liquid change with change in temperature?

Coefficient of viscosity of a liquid rapidly decreases with increase or in its temperature.

Q.10 Application:

It is used to measure the viscosity of liquids, especially in industries like lubricants & pharmaceuticals.